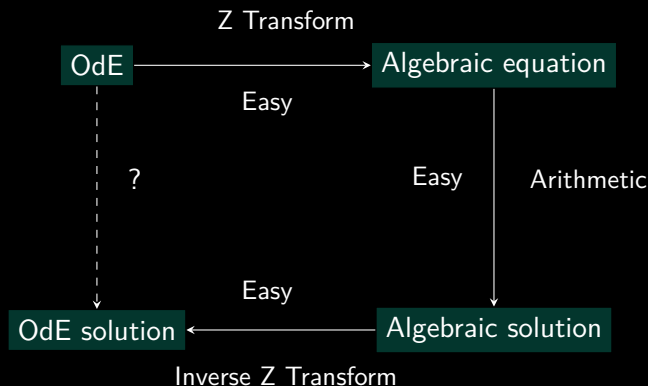


Introduction to Modern Controls

Z transform



The Z transform approach to Ordinary difference Equations (OdEs)



- analogous to Laplace transform for continuous-time signals

Definition

- let $x(k)$ be a real discrete-time sequence that is zero if $k < 0$
- the (one-sided) Z transform of $x(k)$ is

$$\begin{aligned} X(z) &\triangleq \mathcal{Z}\{x(k)\} = \sum_{k=0}^{\infty} x(k)z^{-k} \\ &= x(0) + x(1)z^{-1} + x(2)z^{-2} + \dots \end{aligned}$$

where $z \in \mathbb{C}$

- a linear operator: $\mathcal{Z}\{\alpha f(k) + \beta g(k)\} = \alpha \mathcal{Z}\{f(k)\} + \beta \mathcal{Z}\{g(k)\}$
- the series $1 + \gamma + \gamma^2 + \dots$ converges to $\frac{1}{1-\gamma}$ for $|\gamma| < 1$ [region of convergence (ROC)]
- (also, recall that $\sum_{k=0}^N \gamma^k = \frac{1-\gamma^{N+1}}{1-\gamma}$ if $\gamma \neq 1$)

Example: geometric sequence $\{a^k\}_{k=0}^{\infty}$

$$\sum_{k=0}^{\infty} \gamma^k = \frac{1}{1-\gamma}$$

- $x(k) = a^k$

- $\mathcal{Z}\{a^k\} = \sum_{k=0}^{\infty} a^k z^{-k} = \boxed{\frac{1}{1 - az^{-1}}} = \frac{z}{z-a}$

Example: step sequence (discrete-time unit step function)

$$\mathcal{Z}\{a^k\} = \frac{1}{1 - az^{-1}}$$

- $1(k) = \begin{cases} 1, & \forall k = 0, 1, 2, \dots \\ 0, & \forall k = \dots, -1 \end{cases}$

- $\mathcal{Z}\{1(k)\} = \mathcal{Z}\{a^k\}|_{a=1} = \boxed{\frac{1}{1 - z^{-1}}} = \frac{z}{z-1}$

Example: discrete-time impulse

- $\delta(k) = \begin{cases} 1, & k = 0 \\ 0, & \text{otherwise} \end{cases}$
- $\mathcal{Z}\{\delta(k)\} = 1$

Exercise: $\cos(\omega_0 k)$

$f(k)$	$F(z)$	ROC
$\delta(k)$	1	All z
$a^k 1(k)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$-a^k 1(-k - 1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$ka^k 1(k)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
$-ka^k 1(-k - 1)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
$\cos(\omega_0 k)$	$\frac{1 - z^{-1} \cos(\omega_0)}{1 - 2z^{-1} \cos(\omega_0) + z^{-2}}$	$ z > 1$
$\sin(\omega_0 k)$	$\frac{z^{-1} \sin(\omega_0)}{1 - 2z^{-1} \cos(\omega_0) + z^{-2}}$	$ z > 1$
$a^k \cos(\omega_0 k)$	$\frac{1 - az^{-1} \cos(\omega_0)}{1 - 2az^{-1} \cos(\omega_0) + a^2 z^{-2}}$	$ z > a $
$a^k \sin(\omega_0 k)$	$\frac{az^{-1} \sin(\omega_0)}{1 - 2az^{-1} \cos(\omega_0) + a^2 z^{-2}}$	$ z > a $

Properties of Z transform: time shift

- let $\mathcal{Z}\{x(k)\} = X(z)$ and $x(k) = 0 \ \forall k < 0$
- one-step delay:

$$\begin{aligned}\mathcal{Z}\{x(k-1)\} &= \sum_{k=0}^{\infty} x(k-1)z^{-k} = \sum_{k=1}^{\infty} x(k-1)z^{-k} + x(-1) \\ &= \sum_{k=1}^{\infty} x(k-1)z^{-(k-1)}z^{-1} + x(-1) \\ &= z^{-1}X(z) + \cancel{x(-1)} = \boxed{z^{-1}X(z)}\end{aligned}$$

- analogously, $\mathcal{Z}\{x(k+1)\} = \sum_{k=0}^{\infty} x(k+1)z^{-k} = \boxed{zX(z) - zx(0)}$
- thus, if $x(k+1) = Ax(k) + Bu(k)$ and $x(0) = 0$,

$$zX(z) = AX(z) + BU(z) \Rightarrow X(z) = (zI - A)^{-1}BU(z)$$

provided that $(zI - A)$ is invertible

Solving difference equations

Solve the difference equation

$$y(k) + 3y(k-1) + 2y(k-2) = u(k-2)$$

where $y(-2) = y(-1) = 0$ and $u(k) = 1(k)$.

- $\mathcal{Z}\{y(k-1)\} = z^{-1}\mathcal{Z}\{y(k)\} = z^{-1}Y(z)$
- $\mathcal{Z}\{y(k-2)\} = z^{-1}\mathcal{Z}\{y(k-1)\} = z^{-2}Y(z)$
- $\mathcal{Z}\{u(k-2)\} = z^{-2}U(z)$
- $\Rightarrow (1 + 3z^{-1} + 2z^{-2})Y(z) = z^{-2}U(z)$
- $\Rightarrow \boxed{Y(z) = \frac{z^{-2}}{1 + 3z^{-1} + 2z^{-2}}U(z)}$

Solving difference equations

Solve the difference equation

$$y(k) + 3y(k-1) + 2y(k-2) = u(k-2)$$

where $y(-2) = y(-1) = 0$ and $u(k) = 1(k)$.

- $$Y(z) = \frac{z^{-2}}{1 + 3z^{-1} + 2z^{-2}} U(z) = \frac{z^{-2}}{(1 + 2z^{-1})(1 + z^{-1})} U(z)$$
- $u(k) = 1(k) \Rightarrow U(z) = 1/(1 - z^{-1})$
- $$\Rightarrow Y(z) = \frac{z^{-2}}{(1-z^{-1})(1+2z^{-1})(1+z^{-1})} = \frac{1}{6} \frac{1}{1-z^{-1}} + \frac{1}{3} \frac{1}{1+2z^{-1}} - \frac{1}{2} \frac{1}{1+z^{-1}}$$

(careful with the partial fraction expansion)
- inverse Z transform then gives
$$y(k) = \frac{1}{6}1(k) + \frac{1}{3}(-2)^k - \frac{1}{2}(-1)^k, \quad k \geq 0$$

From difference equation to transfer functions

- general discrete-time OdE:

$$y(k) + a_{n-1}y(k-1) + \cdots + a_0y(k-n) = b_mu(k+m-n) + \cdots + b_0u(k-n)$$

where $y(k) = 0 \ \forall k < 0$

- applying Z transform to the OdE yields

$$(z^n + a_{n-1}z^{n-1} + \cdots + a_0) Y(z) = (b_mz^m + b_{m-1}z^{m-1} + \cdots + b_0) U(z)$$

- hence

$$Y(z) = \underbrace{\frac{b_mz^m + b_{m-1}z^{m-1} \cdots + b_1z + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0}}_{G_{yu}(z): \text{ discrete-time transfer function}} U(z)$$

DC gain of discrete-time transfer functions

- general discrete-time OdE and transfer function:

$$y(k) + a_{n-1}y(k-1) + \cdots + a_0y(k-n) = b_mu(k+m-n) + \cdots + b_0u(k-n)$$

$$Y(z) = \underbrace{\frac{b_m z^m + b_{m-1} z^{m-1} \cdots + b_1 z + b_0}{z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0}}_{G_{yu}(z): \text{ discrete-time transfer function}} U(z)$$

- assuming constant input and convergent output, then at steady state,
 - ▶ $y(k) = y(k-1) = \cdots = y(k-n) \triangleq y_{ss}$ and
 $u(k+m-n) = u(k+m-n-1) = \cdots = u(k-n) \triangleq u_{ss}$
 - ▶ $y_{ss} + a_{n-1}y_{ss} + \cdots + a_0y_{ss} = b_mu_{ss} + \cdots + b_0u_{ss}$
- thus,

$$\underline{\text{DC gain of } G_{yu}(z)} = \frac{b_m + b_{m-1} + \cdots + b_0}{1 + a_{n-1} + \cdots + a_0} = \underline{G_{yu}(z)|_{z=1}}$$

Transfer functions in two domains

$$y(k) + a_{n-1}y(k-1) + \dots + a_0y(k-n) = b_mu(k+m-n) + \dots + b_0u(k-n)$$
$$\iff G_{yu}(z) = \frac{B(z)}{A(z)} = \frac{b_mz^m + b_{m-1}z^{m-1} + \dots + b_1z + b_0}{z^n + a_{n-1}z^{n-1} + \dots + a_1z + a_0}$$

v.s.

$$\frac{d^ny(t)}{dt^n} + a_{n-1}\frac{d^{n-1}y(t)}{dt^{n-1}} + \dots + a_0y(t) = b_m\frac{d^mu(t)}{dt^m} + b_{m-1}\frac{d^{m-1}u(t)}{dt^{m-1}} + \dots + b_0u(t)$$
$$\iff G_{yu}(s) = \frac{B(s)}{A(s)} = \frac{b_ms^m + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$$

Properties	$G_{yu}(s)$	$G_{yu}(z)$
poles and zeros	roots of $A(s)$ and $B(s)$	roots of $A(z)$ and $B(z)$
causality condition	$n \geq m$	$n \geq m$
DC gain / steady-state response to unit step	$G_{yu}(0)$	$G_{yu}(1)$

Coding a discrete-time transfer function

```
num = [0.09952, -0.08144];  
den = [1, -1.792, 0.8187];  
Ts = 0.1;  
sys_tf = tf(num,den,Ts)  
poles = pole(sys_tf);  
zeros = zero(sys_tf);  
disp(['System Poles = ',num2str(poles)])  
disp(['System Zeros = ',num2str(zeros)])  
  
[yout, T] = step(sys_tf);  
figure, stairs(T, yout)  
figure, impulse(sys_tf)  
  
u1 = 2*ones(length(T),1);  
u2 = sin(T);  
figure, lsim(sys_tf,u1,T)  
figure, lsim(sys_tf,u2,T)
```

```

import control as co
import matplotlib.pyplot as plt
import numpy as np
Ts = 0.1 # sampling time
num = [0.09952, -0.08144] # Numerator co-efficients
den = [1, -1.792, 0.8187] # Denominator co-efficients
sys_tf = co.tf(num,den, Ts)
print(sys_tf)

poles = co.poles(sys_tf)
zeros = co.zeros(sys_tf)
print('\nSystem Poles = ', poles, '\nSystem Zeros = ', zeros)

T,yout = co.step_response(sys_tf)
plt.figure(1,figsize = (6,4))
plt.step(T,np.append(0,yout[0:-1]))
plt.grid(True)
plt.ylabel("y")
plt.xlabel("Time (sec)")
plt.show()

```

```
import control as co
import matplotlib.pyplot as plt
import numpy as np
Ts = 0.1 # sampling time
num = [0.09952, -0.08144] # Numerator co-efficients
den = [1, -1.792, 0.8187] # Denominator co-efficients
sys_tf = co.tf(num,den, Ts)
print(sys_tf)

T,yout_i = co.impulse_response(sys_tf)
plt.figure(1,figsize = (6,4))
plt.step(T,np.append(0,yout_i[0:-1]))
plt.grid(True)
plt.ylabel("y")
plt.xlabel("Time (sec)")
plt.show()
```

Additional useful properties of Z transform

- time shifting (assuming $x(k) = 0$ if $k < 0$):

$$\mathcal{Z} \{x(k - n_d)\} = z^{-n_d} X(z)$$

- Z-domain scaling: $\mathcal{Z} \{a^k x(k)\} = X(a^{-1}z)$
- differentiation: $\mathcal{Z} \{kx(k)\} = -z \frac{dX(z)}{dz}$
- time reversal: $\mathcal{Z} \{x(-k)\} = X(z^{-1})$
- convolution: let $f(k) * g(k) \triangleq \sum_{j=0}^k f(k-j) g(j)$, then

$$\mathcal{Z} \{f(k) * g(k)\} = F(z) G(z)$$

- initial value theorem: $f(0) = \lim_{z \rightarrow \infty} F(z)$
- final value theorem: $\lim_{k \rightarrow \infty} f(k) = \lim_{z \rightarrow 1} (z-1) F(z)$, if $\lim_{k \rightarrow \infty} f(k)$ exists and is finite

Mortgage payment

- imagine you borrow \$100,000 (e.g., for a mortgage)
- annual percent rate: $APR = 4.0\%$
- plan to pay off in 30 years with fixed monthly payments
- interest computed monthly
- what is your monthly payment?

Mortgage payment

- borrow \$100,000 $\Rightarrow$ initial debt $y(0) = 100,000$
- $APR = 4.0\% \Rightarrow MPR = \frac{4.0\%}{12} = 0.0033$
- pay off in 30 years ($N = 30 \times 12 = 360$ months) $\Rightarrow y(N) = 0$
- debt at month $k + 1$:

$$y(k+1) = \underbrace{(1 + MPR)}_a y(k) - \underbrace{b}_{\text{monthly payment}} 1(k)$$

- $\Rightarrow Y(z) = \frac{z}{z-a} y(0) - \frac{1}{z-a} \frac{b}{1-z^{-1}}$
- $\Rightarrow Y(z) = \frac{1}{1-az^{-1}} y(0) + \frac{b}{1-a} \left(\frac{1}{1-az^{-1}} - \frac{1}{1-z^{-1}} \right)$
- $\Rightarrow y(k) = a^k y(0) + \frac{b}{1-a} (a^k - 1)$
- need $y(N) = 0 \Rightarrow a^N y(0) = -\frac{b}{1-a} (a^N - 1)$
- $\Rightarrow b = \frac{a^N y(0)(a-1)}{a^N - 1} = \477.42